

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Compare L1, L2 and L3 Cache

L1, L2 and L3 are cache levels used to reduce the time required to access frequently used data.

Feature	L1 Cache	L2 Cache	L3 Cache
Size	Smallest	Medium	Largest
Latency	Lowest	Low	Higher than L1/L2
Bandwidth	Highest	High	Lower than L1/L2
Access spe	Fastest	Fast	Slower than L1/L2
Location	Inside CPU core	Usually near/inside core	Usually shared among cor
Purpose	Immediate data/instructions	Secondary cache	Shared data cache

Impact on Processor Performance

- L1 cache provides very fast access to frequently used data.
- L2 cache stores data that is not found in L1.
- L3 cache provides a larger shared storage area and reduces accesses to main memory.

- A higher cache hit rate reduces processor waiting time.
- Therefore, the hierarchy $L1 \rightarrow L2 \rightarrow L3 \rightarrow \text{Main Memory}$ improves overall processor performance.

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

Compare SRAM and DRAM

SRAM and DRAM are both volatile memory technologies but are used differently in a computer's memory hierarchy.

Feature	SRAM	DRAM
Full form	Static RAM	Dynamic RAM
Latency	Very low	Higher than SRAM
Bandwidth	High	High
Throughput	High for cache access	High for main-memory access
Refresh required	No	Yes
Cost	Expensive	Cheaper
Density	Lower	Higher
Main use	Cache memory	Main memory

SRAM

SRAM uses flip-flop-based storage and does not require periodic refreshing. It provides very low latency, making it suitable for L1, L2 and L3 cache.

DRAM

DRAM stores data using capacitors and requires periodic refreshing. It has higher latency than SRAM but provides greater density at lower cost, making it suitable for main memory.

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Compare Virtual Memory, Paging and Cache Memory

Virtual memory, paging and cache memory have different purposes but work together to improve system performance.

Feature	Virtual Memory	Paging	Cache Memory
Purpose	Extends memory using storage	Manages virtual-to-physical memory	Speeds up CPU access
Access latency	High	Higher than cache	Very low
Storage	RAM + secondary storage	Pages in RAM/storage	SRAM
Speed	Slow compared with RAM/cache	Relatively slow	Fastest
Throughput	Lower	Moderate	High
Managed by	OS	OS/MMU	Hardware

Virtual Memory

Virtual memory allows programs to use an address space larger than the available physical RAM. When required data is not in RAM, it may need to be obtained from secondary storage, increasing access latency.

Paging

Paging divides memory into fixed-size pages and physical memory into frames. **It** allows efficient management of virtual memory.

Cache Memory

Cache stores frequently accessed instructions and data close to the processor. It significantly reduces CPU access latency.

Impact on Throughput

Cache improves throughput by reducing CPU waiting time. Paging and virtual memory allow efficient memory management but can reduce performance when frequent page transfers occur.

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

Compare NAND Flash and DRAM

NAND Flash and DRAM have different characteristics and therefore occupy different positions in the memory hierarchy.

Feature	NAND Flash	DRAM
Type	Non-volatile	Volatile
Latency	Higher	Lower
Bandwidth	Moderate to high	High
Throughput	Good for storage	High for memory access
Refresh	Not required	Required
Power	Low when idle	Higher
Persistence	Yes	No
Main use	SSD/storage	Main memory

NAND Flash

- NAND Flash retains data even when power is removed. It is commonly used in SSDs and other storage devices.
- Its access latency is much higher than DRAM, so it is not normally used as the main working memory of the CPU.

DRAM

- DRAM provides much faster access and is used as the main memory between cache and secondary storage.

Suitability in Hierarchical Memory

A typical hierarchy is:

CPU → Cache → DRAM → NAND Flash

- Cache: fastest
- DRAM: fast working memory
- NAND Flash: persistent high-capacity storage

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Compare MRAM and RRAM

MRAM and RRAM are advanced non-volatile memory technologies designed to provide fast access while retaining data without power.

Feature	MRAM	RRAM
Full form	Magnetoresistive RAM	Resistive RAM
Data storage	Magnetic state	Resistance state
Volatile/Non-volatile	Non-volatile	Non-volatile
Access time	Very fast	Fast
Throughput	High	High
Energy efficiency	High	High
Endurance	High	Potentially high
Main advantage	Fast and durable	High density and scalability

MRAM

MRAM stores information using magnetic states. It provides fast access, good endurance and non-volatility. It can be useful where fast memory access and low standby power are required.

RRAM

RRAM stores information by changing the resistance of a material. It has potential for high density and low-power memory applications.

Performance Comparison

Access Time:

MRAM provides very fast access. RRAM can also provide fast access, depending on the technology and implementation.

Throughput:

Both technologies can provide high throughput and are candidates for future memory systems.

Energy Efficiency:

Both are non-volatile, so they can reduce energy associated with maintaining stored data when compared with volatile memories.

Code of Conduct:

*I (AMARA SATWIKA / 192525204) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: A.SATWIKA

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand